

HOPPER INFORMATION SERIES

03 / BLOCK CODING

Coding blocks reference

What each Hopper Studio block category does, what it returns and how commands sequence.

FLIGHT

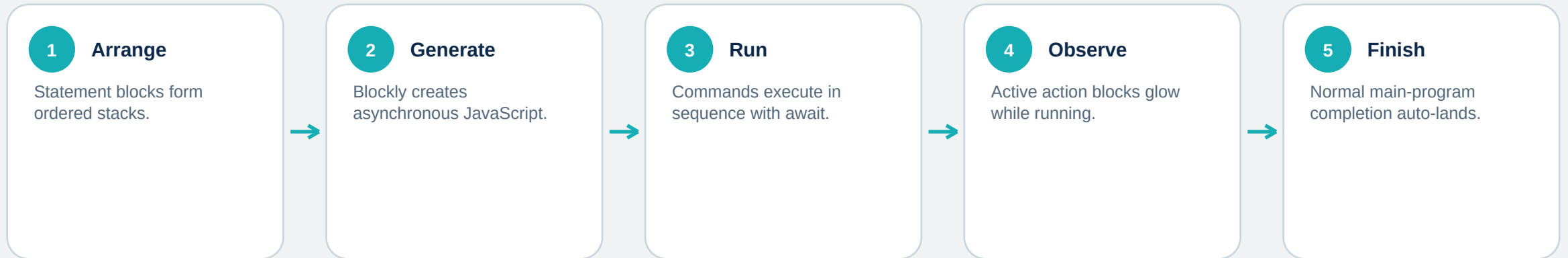
VISION

LOGIC + LOOPS





How a block program becomes a flight



when program starts

take off

fly forward for 1 sec at 15%

land

GENERATED JAVASCRIPT

```
1  await runtime.runBlock("takeoff", async () => {
2    await drone.takeOff();
3  });
4  await runtime.runBlock("fly-1", async () => {
5    await drone.fly("forward", 1, 15);
6  });
7  await runtime.runBlock("land", async () => {
8    await drone.land();
9  });
```



Start & Events

when program starts

Container for the main stack. Other top-level stacks can also be generated.

stop program

Ends execution but does NOT send a land command. Land first or use UI STOP & LAND.

wait [seconds]

Pauses in sequence. Negative/coercion failures become zero.

print [value]

Writes a line to the in-app program console.

when [key] is [pressed/released]

Arrow, Space and a-z events. Event programs keep listening until Stop.

[key] key is pressed

Boolean from the live pressed-key set.

continue if [condition]

False returns from the current main, event or function.

SAFETY

The red UI STOP & LAND is the dependable way to cancel tasks and send a landing command.



Mini Drone: flight blocks

take off / land / hover

Awaited state-changing actions.

fly [direction] for [s] at [power]%

6 directions; time ≥ 0 ; teach 0-100% power; default 1 s / 15%.

rotate [degrees] [direction]

Timed/open-loop yaw at about 180 deg/s, then settle.

flip [direction]

Forward/back/left/right; preserve safe height and clear space.

set [axis] to [power]%

Persistent pitch/roll/yaw/gaz power until reset. 'Altitude' means gaz, not height.

reset movement

Zeros all motion axes; does not land.

center on AprilTag

Repeated scans, translation pulses and yaw alignment; does not control height or land.

cut off motors

Emergency only. Immediate; block/code has no confirmation.

Fly adds a stabilization wait after motion. set-axis does not.



Mini Drone: state, events and accessories

SENSORS + EVENTS

- battery level -> number 0-100 or null before telemetry.
- drone is flying / landed -> Boolean.
- wait until battery changes -> asynchronous wait.
- when drone starts flying / lands / crashes / battery changes.

ACCESSORIES

- take and store photo -> session gallery JPEG from camera pixels.
- open / close grabber -> needs physical claw; simulator is a timing stub.
- fire cannon -> needs physical cannon; simulator is a timing stub.

NOT AVAILABLE AS BLOCK VALUES

GPS | x/y room position | real altitude | velocity | attitude | heading | raw IMU | range

'set altitude' is signed vertical motor power. It is not an altitude setpoint.



Camera Vision blocks

BINARY

sees color: threshold + coverage

center pixel color: threshold

OBJECTS

scan objects (default 55%)

sees label at confidence %

last label x/y at confidence %

CUSTOM

sees custom label at confidence %

APRILTAGS

scan AprilTags

sees tag ID any / 0-586

center tag: power + slack + lost

Binary: threshold + coverage are 0-100%; invert swaps classes; predicates await a fresh frame and return Boolean.

COCO: label + confidence + saved coordinate (-100..100, or 0). Custom: label + confidence only; no box or coordinate.

AprilTags: scan returns detections; sees/center return Boolean; center also stops at 30 s or the lost-search limit.



Logic, loops, math, variables and functions

LOGIC

- if / else
- comparisons
- and / or / not
- true / false
- ternary

LOOPS

- forever
- for N seconds
- repeat count
- while / until
- counted for
- break / continue

MATH

- arithmetic
- unary math
- trig
- rounding
- modulo
- random

VARIABLES

- create
- get
- set
- change

FUNCTIONS

- define
- parameters
- return value
- awaited calls

Loops yield so Stop can interrupt. Generated custom functions are async; calls are awaited.



Build one intentional, safe mission

when program starts

take off

set found to false

repeat for 10 seconds

if camera sees white paper

set found to true

break out of loop

else: fly forward 0.5 sec at 12%

land after the loop

MISSION RULES

- Place all intended executable stacks deliberately; detached top-level stacks may still run.
- Use one start hat for the main mission.
- Break after the target is found; keep land outside the loop so every normal path lands.
- Use the UI STOP & LAND for intervention.

SOURCES AND FURTHER READING

Hopper Studio block definitions and JavaScript generators

lib/blockly.ts in the Hopper Studio project

Hopper Studio execution and safety behavior

components/HopperStudio.tsx, lib/runtime.ts and README.md